

Lesson 22: Geometric Power Series

Rubric: Geometric Power Series Representation

Criteria	4 (Exceeds)	3 (Meets)	2 (Partially Meets)	1 (Below Standard)
Identification & Setup	Identifies the correct function form $f(x) = \frac{a}{1-r}$, correctly identifying first term a and ratio r (including signs) without errors.	Identifies the correct formula with minor algebraic errors in recognizing a or r.	Incorrectly identifies a or r due to significant algebraic errors.	Cannot identify the function as a geometric series.
Series Generation	Correctly writes the series $\sum ar^n$ using proper summation notation, including the starting index n.	Writes the correct series but fails to use proper sigma notation.	Creates the series with significant errors in a or r.	Incorrect series generated.
Interval of Convergence (IoC)	Correctly finds the interval of convergence using $ r < 1$, including checking endpoints.	Finds the radius of convergence but makes errors determining the interval bounds.	Attempts to find the interval but fails to use absolute value.	Incorrect convergence interval.

Algebraic Manipulation	Correctly manipulates the function into $a/(1-r)$ form (e.g., factoring out constants) without errors.	Manipulates the function with minor errors in constants.	Struggles to manipulate the function, leading to an incorrect r .	Cannot transform the function into a geometric form.
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